Investigating Binary Coded Excitation Schemes for US Wearables with Low-Complexity Electronics

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Background, Motivation and Objective

Ultrasound (US) wearables enable a multitude of novel applications in clinical, lifestyle, and rehabilitation fields by their ability to continuously monitor biomechanical properties over long periods of time. In comparison to high-end clinical US devices, such miniaturized systems are constrained by power consumption, computational resources, and hardware complexity, making it difficult to maintain high signal quality at low transmit energy. To improve the signal quality and address the trade-off between signal-to-noise ratio (SNR) and energy efficiency, we investigated binary-coded excitation combined with pulse compression (PC) on a resource-constrained microcontroller platform.

Statement of Contribution/Methods

We evaluated different binary coding strategies, specifically Barker codes, Golay codes, and Maximum Length Sequences (MLS), through numerical studies comparing acquisition methods and assessing matched-filter reconstruction with respect to axial resolution and detectability. Experimental validation was then performed using a 7MHz piezoelectric TX/RX pair and a planar reflector in water to verify code-based gain and main lobe narrowing relative to short pulses. An implementation on a PSoC Edge microcontroller demonstrated the feasibility of coded excitation and PC under wearable-class constraints. A central feature is the hardware-oriented MLS generation scheme in which the sequence is produced entirely by an integrated feedback shift register. This removes sequence generation from the CPU and enables a timing-deterministic excitation suited to an energy-constrained wearable ultrasound operation.

Results/Discussion

While Barker codes showed limitations at higher measurement depths, both MLS and Golay codes provided robust signal reconstruction when using periodic signals. Based on these findings, MLS was selected for implementation due to its reduced complexity in hardware integration. Experimental results validate this approach, demonstrating approximately 25% narrower compression peaks and a modest SNR improvement of 5dB relative to a standard pulse-echo reference. These findings confirm that embedded binary-coded excitation is a viable strategy for enabling high-performance, energy-efficient US wearables.

a) b)

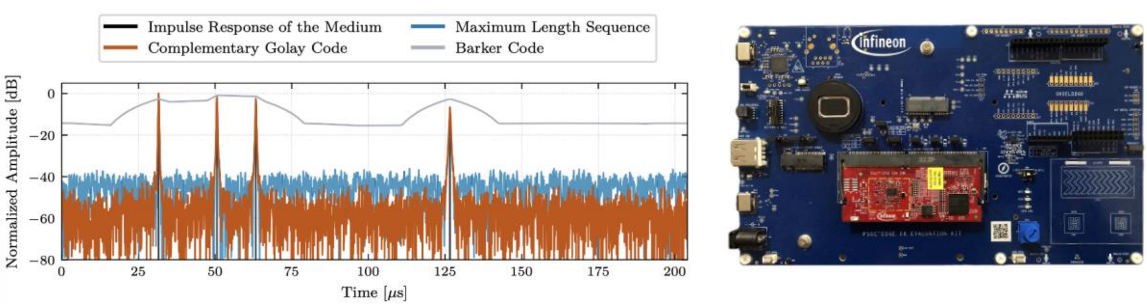

Fig. 1: a) PC results from numerical studies, b) microcontroller board used for experiments